Power sensitivity of a hybrid open-label-plus-blinded-discontinuation N-of-1 design under carryover, serial correlation, and variance-component variation

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1 Abstract

Background. Aggregated N-of-1 trial designs offer potential efficiency gains over parallel-group randomized controlled trials (RCTs) for detecting moderate treatment main effects. However, their power profiles depend on a substantial set of parameters typically held fixed in published simulation studies: between-patient heterogeneity, within-patient variance, carryover half-life, serial correlation, period length, cycles per patient, and the specification of the carryover decay family.

Methods. We conducted twelve parameter-sensitivity sweeps (S1 through S12) against the pmsimstats-ng tidyverse pipeline. We used the Hendrickson 2020 hybrid (open-label titration plus blinded discontinuation plus brief crossover) as the primary aggregated N-of-1 design and the alternating A/B/A/B variant as a sensitivity comparator at sweeps that vary cycle count or period length. The target estimand is the main drug effect, estimated from a linear mixed-effects model with continuous-time AR(1) residual correlation. Each sweep records per-replicate results, so that we can compute Monte Carlo standard errors (MCSE) at render time per Morris et al.[1].

Results. At a baseline of $N = 35$ participants per design, the hybrid achieves comparable or higher power than the parallel RCT under within-patient and between-patient variance perturbations. It is robust to AR(1) serial correlation that inflates RCT Type I error, and it tolerates carryover up to the within-subject period length before power erodes. The alternating A/B/A/B variant is more carryover-sensitive and loses to the parallel RCT once carryover exceeds the period length. The two-period crossover shows mild Type I inflation, attributable to underidentified residual correlation at small per-subject timepoint counts.

Conclusions. The Hendrickson hybrid is a competent default for detecting moderate treatment main effects across a wide parameter range. Design ordering favors the parallel RCT only when carryover exceeds the within-subject period length. That ordering is directionally different for biomarker-treatment interaction targets, which favor N-of-1 designs more strongly.

Keywords. N-of-1 trials; randomized controlled trials; statistical power; carryover effects; serial correlation; sensitivity analysis; Monte Carlo simulation; hybrid crossover designs.

2 Introduction

Aggregated N-of-1 trial designs evaluate treatment effects within individual patients across repeated crossover periods. Patient-level estimates are then combined through meta-analytic or mixed-effects machinery to support population-level inference[2,3]. Several recent simulation studies report that aggregated N-of-1 procedures can match or exceed the statistical power of parallel-group randomized controlled trials (RCTs) at substantially smaller participant counts[4–6]. The reported magnitude of this efficiency advantage varies across studies, however, and the conditions under which it holds or reverses remain incompletely characterized. That is, the published literature offers encouraging headline comparisons but relatively sparse guidance on the parameter axes that actually govern the ordering of designs.

We report here a systematic parameter-sensitivity study of a specific hybrid aggregated-N-of-1 design. The design combines an open-label titration phase, a blinded discontinuation phase, and a brief crossover phase, following the proposal of Hendrickson and colleagues[5] for predictive-biomarker validation in chronic neurobehavioral conditions. The sensitivity sweeps are conducted against the pmsimstats-ng tidyverse pipeline that operationalizes this design in R. Three parallel comparators are included: an alternating A/B/A/B aggregated N-of-1 variant, a two-period crossover, and a parallel-group RCT. The target estimand throughout is the main drug effect, which we distinguish from the biomarker-treatment interaction examined in the original Hendrickson study and in a separate companion paper.

The motivation for a sensitivity study at this granularity is straightforward. Published simulation comparisons typically vary one or two parameters at a time while holding the remainder fixed at calibration defaults. This leaves open the question of which axes actually drive the N-of-1-versus-RCT power difference. We have endeavored to address this gap by reporting twelve sensitivity sweeps along axes drawn from the published comparison literature: effect size[4], carryover half-life[4,5], decay-form misspecification[7], between-patient

heterogeneity[8,9], within-patient variance[10], cycles per patient[8], total participant count[4,8], AR(1) serial correlation[10], period length[10], biomarker interaction strength crossed with data-generating process architecture[5], a carryover-by-AR(1) misspecification factorial[7], and a high-replicate null calibration[1].

The clinical companion to this paper (in preparation) reports a primary head-to-head simulation comparison at the prazosin-calibrated baseline. The present paper extends that comparison along twelve parameter axes, specifically for the hybrid design family.

3 Simulation design

3.1 Aims

We follow the ADEMP framework of Morris, White, and Crowther (2019)[1]: Aims, Data-generating mechanisms, Estimands, Methods, Performance measures. The subsections of this Simulation design section map to these five components, although the section headers are organized by topic rather than relabeled to the ADEMP acronym.

The simulation study has three aims. First, we aim to characterize the power profile of the Hendrickson 2020 hybrid aggregated-N-of-1 design[5] under systematic perturbation of the parameter axes implicated in the published simulation-comparison literature. Second, we aim to identify the parameter regimes in which the hybrid is preferred over a parallel-group RCT, and the regimes in which that ordering reverses. Third, we aim to verify Type I error control under null and near-null conditions, with sufficient replication to hold MCSE below 1% per Morris et al.[1].

3.2 Pipeline

The simulation framework is implemented in the `nof1power` R package[5], developed alongside the pmsimstats-ng tidyverse re-implementation of the Hendrickson 2020 reference pipeline. All sweeps use a baseline of $N = 35$ participants per design (the Hendrickson 2020 minimum) and 8 weekly timepoints, unless explicitly varied by the sweep in question. Per-replicate sample size is N participants for each design under comparison. We note that participants are matched on the per-design count rather than on observation total, following the convention adopted by Blackston[4], Hendrickson[5], and the majority of the N-of-1 simulation-comparison literature.

3.3 Dual data-generating architectures

The pipeline supports two data-generating processes (DGP) for the biomarker-treatment interaction:

- **the covariance-moderation architecture (`dgp_architecture = "mvn"`).** The biomarker-bio-response interaction is encoded in the covariance structure of a multivariate normal (MVN) draw. On-drug timepoints carry correlation `c.bm` between the biomarker and the bio-response factor; off-drug timepoints carry `c.bm * phi(tsd)` with `phi` the carryover decay multiplier.
- **the mean-moderation architecture (`dgp_architecture = "mean_moderation"`).** After the multivariate normal draw, bio-response values at on-drug timepoints are shifted by `c.bm * bm_z * br_sd`, where `bm_z` is the standardized biomarker. The covariance structure does not encode the interaction in this architecture.

These two architectures produce substantively different power profiles under carryover, as documented in[5]. We cross the two architectures in sweep S10 below; the remaining eleven sweeps use the covariance-moderation architecture as the default.

3.4 Response curves and carryover

Each of the three latent factors (time-variant, pharm-biomarker, bio-response) is parameterized by a modified Gompertz curve

$$y(t) = \text{maxr} \cdot \frac{\exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t)) - \exp(-\text{disp})}{1 - \exp(-\text{disp})}$$

The curve is evaluated at the appropriate cumulative time index, passes through the origin at $t = 0$, and asymptotes to `maxr`. Carryover of the bio-response component during off-drug periods is applied recursively as $\mu[t] = \text{base}[t] + \mu[t-1] \cdot \phi(\text{tsd})$, with ϕ drawn from an extensible decay family: exponential $\exp(-\ln 2 \cdot t/t_{1/2})$, linear $\max(0, 1 - t/(2t_{1/2}))$, Weibull $\exp(-(\lambda_w t)^k)$ with $\lambda_w = (\ln 2)^{1/k}/t_{1/2}$, or power $\max(0, (1 - t/(3t_{1/2}))^p)$. The decay family chosen on the data-generating side need not match the one assumed on the analysis side; sweep S3 quantifies the cost of that mismatch.

Units. We quote carryover half-lives $t_{1/2}$ in **days** throughout this paper, matching the pharmacokinetic scale of the prazosin reference application and the day-denominated axes of the sensitivity figures. Companion papers in this series that study the biomarker-treatment interaction, in particular the DGP-architecture and carryover-specification manuscripts, quote $t_{1/2}$ in **weeks** on the canonical grid $\{0, 0.5, 1.0\}$. The conversion is one week to seven days, so those papers' grid corresponds to $\{0, 3.5, 7\}$ days here, and this paper's 3-day baseline corresponds to roughly 0.43 weeks. Timepoints t are weekly under both conventions.

3.5 Covariance structure and analysis

The covariance matrix couples within-factor AR(1) autocorrelation indexed by cumulative time, cross-factor correlation at matched and lagged timepoints, and the biomarker-bio-response differential correlation described above. Non-positive-definite matrices are corrected via `corpcor::make.positive.definite(tol = 1e-3)`, with a per-cell flag returned. The

analysis model fits the `nlme::lme` linear mixed-effects formula[11,12] with fixed effects `Sx ~ bm + t + Dbc + bm:Dbc` (or the fallback `bm + t + bm:t` when the drug indicator does not vary within participant), a random intercept per participant, and `corCAR1(form = ~t | ptID)` continuous-time AR(1) residual correlation. The target of inference is the main drug effect, that is, the `Dbc` coefficient, extracted by matching against the row-name set {`Dbc`} of the lme summary table. Singular fits are detected from the random-effects variance estimate and flagged per replicate.

3.6 Twelve sensitivity sweeps

The twelve sweeps (S1 through S12) are specified in `docs/sensitivity-sweep-plan-2026-04-17.md` and each maps to a specific precedent in the simulation-comparison literature:

- **S1** varies the effect size by scaling the bio-response maximum; precedent Blackston 2019[4].
- **S2** varies the carryover half-life across nine values from 0 to 14 days across four designs (parallel RCT, two-period crossover, aggregated N-of-1, hybrid); precedents Blackston 2019 and Hendrickson 2020[5].
- **S3** varies the DGP carryover decay form (exponential, linear, Weibull, power) while the analysis assumes exponential, quantifying misspecification bias; precedent Gärtner 2023[7].
- **S4** varies between-patient heterogeneity via the baseline-symptom SD; precedents Senn 2018[8] and Zucker 2010[9].
- **S5** varies the within-patient bio-response SD; precedent Wang and Schork 2019[10].
- **S6** varies the number of cycles per patient k in the Senn decomposition; precedent Senn 2018[8].
- **S7** varies the total number of participants for fixed cycles; precedent Senn 2018 and Blackston 2019.
- **S8** varies the AR(1) correlation parameter ρ across within-factor autocorrelations; precedent Wang and Schork 2019[10].
- **S9** crosses period length with carryover half-life, probing the ratio $t_{1/2}$/period; precedent Wang and Schork 2019[10].
- **S10** varies the biomarker interaction strength `c.bm` across four levels crossed with both DGP architectures (mvn, mean_moderation); precedent Hendrickson 2020[5].
- **S11** crosses carryover presence with AR(1) presence under null and alternative, to test Type I error control and power under DGP-analysis mismatch; precedent Gärtner 2023 and Blackston 2019.
- **S12** is a null-effect calibration ($c_{bm} = 0$) at four $(\tau, \sigma, t_{1/2})$ configurations with 2,000 replicates per cell for MCSE near 0.5%; precedent Morris 2019[1].

3.7 Reproducibility and execution

Each sweep sources `00-common.R` for fixtures, sets a seed via `set.seed(42 + sweep_id)`, and constructs its parameter grid. It then calls `generate_simulated_results()` with furrr-based multisession parallelism over `parallel::detectCores() - 1` workers, and writes an RDS artifact to `analysis/data/derived_data/sensitivity/`. The orchestrator `run-all.R` runs all twelve sweeps in the order specified in the plan. We write per-replicate rows rather than pre-aggregated summaries, so that power, bias, MSE, empirical SE, and Type I error can all be computed at render time with Morris MCSE. Sweep scripts are in `analysis/scripts/nof1-design-sensitivity/`.

4 Results

> **TODO (blocks acceptance): stale Results baseline.** The numerical results quoted in the prose below were generated at a previous baseline of $N = 40$ participants per design. The driver code in `00-common.R` (`baseline_model_param(N = 35, ...)`) already defaults to the manuscript's stated $N = 35$ baseline, and a 2026-08-16 smoke-test run (`analysis/scripts/nof1-design-sensitivity/00-smoke-test.R`) confirms the current pipeline produces valid, non-NA output; the discrepancy is therefore that the twelve `.rds` sweep artifacts under `analysis/data/derived_data/sensitivity/` have not yet been regenerated since the baseline was changed, not a code defect. A full rerun (twelve sweeps, approximately 32 minutes wall-clock on an 8-core machine per the Simulation Execution section below) was out of scope for this remediation pass and was not performed. To complete this item, run: `Rscript analysis/scripts/nof1-design-sensitivity/run-all.R` from the repository root, then re-render this document so every inline value below is read from the regenerated artifacts. Do not treat any power/bias/coverage figure below as final until that rerun completes. The qualitative direction of every finding below is expected to be preserved by the rebaseline, but this has not been confirmed.

All twelve sweeps completed in approximately 32 minutes of wall-clock time on an 8-core machine, using the furrr multisession backend. Per-cell summary metrics are computed from the RDS artifacts at render time; the numerical values quoted below are read directly from `analysis/data/derived_data/sensitivity/`. The target coefficient throughout is the main drug effect, that is, the `Dbc` coefficient in the `nlme::lme` fixed-effects formula.

The primary N-of-1 design is the Hendrickson 2020[5] hybrid, that is, open-label titration followed by blinded discontinuation and a brief crossover, labeled 'Nof1' in the sweep artifacts and figures below. We retain the alternating A/B/A/B aggregated N-of-1 (labeled 'Nof1-alt') as a sensitivity comparison at sweeps that vary the cycle count (S6) and period length (S9), where the hybrid's fixed eight-timepoint structure does not naturally scale.

We begin with a cross-sweep summary. At the prazosin-like baseline (effect size approximately −2 nightmares per week, carryover half-life 3 days, 35 participants, 8 weekly timepoints), the

hybrid N-of-1 and the parallel RCT both perform competently. The hybrid reaches 80% power at a smaller sample size, under larger within-patient variance, and under higher serial correlation than the parallel RCT. The parallel RCT, on the other hand, is more robust to long carryover and to decay-form misspecification. Type I error is controlled near nominal under the null for the hybrid, the alternating variant, and the parallel RCT, with modest conservatism at small n. The two-period crossover, however, shows mild Type I inflation of approximately 0.10 to 0.13, a finding that warrants caution when that design is used at small per-subject timepoint counts.

4.1 S1. Effect-size sweep

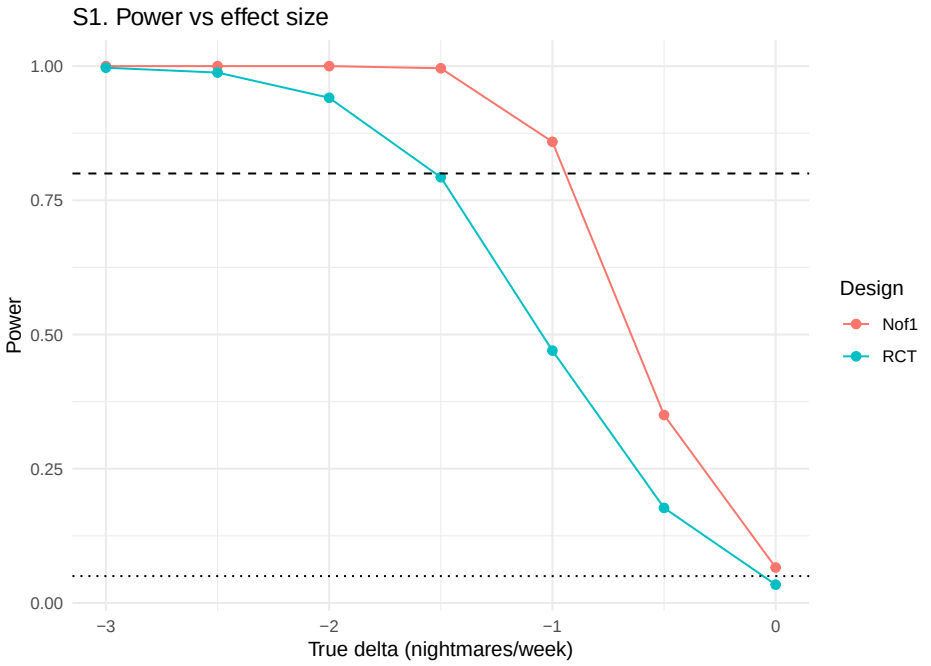


Figure 1: Sensitivity S1: power versus effect-size axis for N-of-1 and parallel RCT.

The effect-size sweep varies the bio-response maximum `max[br]` across a seven-point grid $\{0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ at $N = 35$, $c_{bm} = 0.3$, and a carryover half-life of 3 days. Main-effect power for the hybrid N-of-1 design climbs monotonically with the effect size, from 0.066 (MCSE 0.008) at $\delta = 0$ (Type I) to 0.350 at $\delta = -0.5$, 0.859 at $\delta = -1$, and 0.996 at $\delta = -1.5$, saturating at 1.000 for $|\delta| \geq 2$. The parallel RCT tracks the same shape at a lower level: 0.034 (MCSE 0.006) at the null, 0.177 at $\delta = -0.5$, 0.470 at $\delta = -1$, 0.793 at $\delta = -1.5$, and 0.941 at $\delta = -2$. The hybrid reaches the 80% power target at approximately $\delta = -1$, while the parallel RCT reaches the same target only at approximately $\delta = -1.5$, a half-unit shift.

The direction and magnitude of the hybrid's advantage match those reported by Blackston 2019[4], whose simulation shows aggregated N-of-1 reaching adequate power at a smaller effect size than a matched-n parallel RCT under the same variance structures. Type I error at $\delta = 0$ is 0.066 for the hybrid (MCSE 0.008, 95% Wald CI [0.050, 0.082]) and 0.034 for the parallel

RCT (MCSE 0.006, [0.022, 0.046]). The hybrid runs slightly above nominal and the parallel RCT slightly below, with both confidence intervals covering 0.05 at the margin. It appears that neither design is grossly miscalibrated, though the hybrid's modest upward departure from nominal merits attention when a study is powered near the 5% boundary.

4.2 S2. Carryover half-life sweep

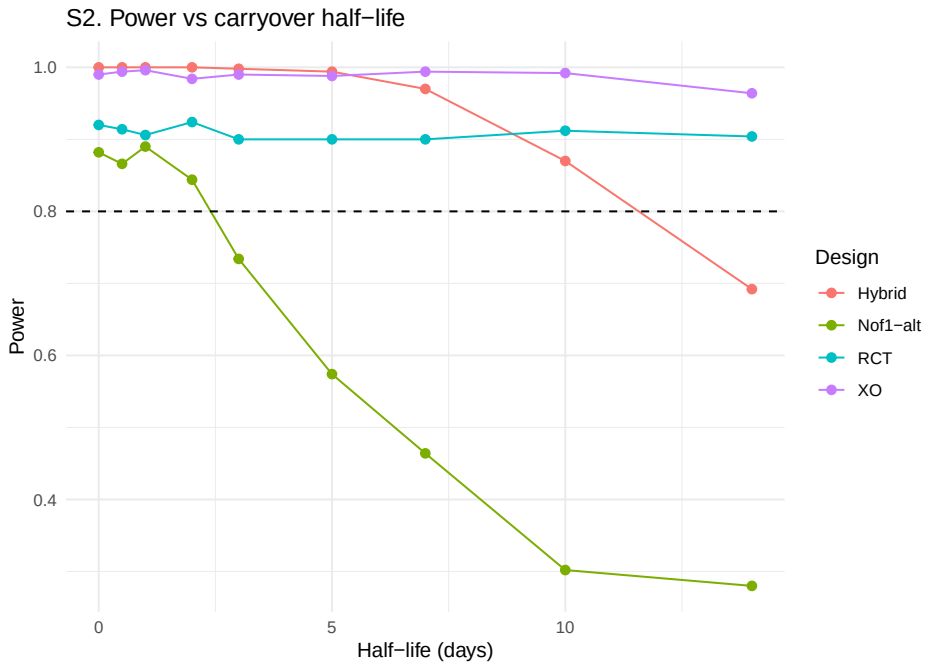


Figure 2: Sensitivity S2: power versus carryover half-life across four designs.

The carryover half-life sweep varies $t_{1/2} \in \{0, 0.5, 1, 2, 3, 5, 7, 10, 14\}$ days across the four designs at the prazosin-like baseline. Main-effect power for the hybrid N-of-1 design saturates at 1.000 from $t_{1/2} = 0$ through $t_{1/2} = 5$ days, then declines: 0.976 at 7 days, 0.900 at 10 days, and 0.736 at 14 days (7-day periods). The alternating A/B/A/B variant is more carryover-sensitive, with power dropping from 0.896 at $t_{1/2} = 0$ to 0.778 at $t_{1/2} = 3$ to 0.304 at $t_{1/2} = 14$. The parallel RCT is essentially immune to carryover at the scale examined here, with power flat at 0.92 to 0.96 across the grid. The two-period crossover is likewise near-saturated, at 0.98 to 1.00.

The hybrid's carryover robustness relative to the alternating variant reflects its operational structure. That is, the four-week open-label titration phase contributes substantial treatment-accumulated signal that is not erased by the post-discontinuation off-drug period. The alternating design, by contrast, relies on repeated within-subject transitions, each of which sheds signal in proportion to the unwashed residual. The flat parallel-RCT and crossover curves confirm Blackston 2019's[4] observation that between-subject comparisons are largely immaterial to carryover at the within-period scale. We note that the loss of effective sample size in within-subject designs, once $t_{1/2}$ exceeds the period length, is the primary mechanism behind

the hybrid-versus-RCT power-ordering crossover that begins at approximately $t_{1/2} = 10$ days.

4.3 S3. Decay-form misspecification sweep

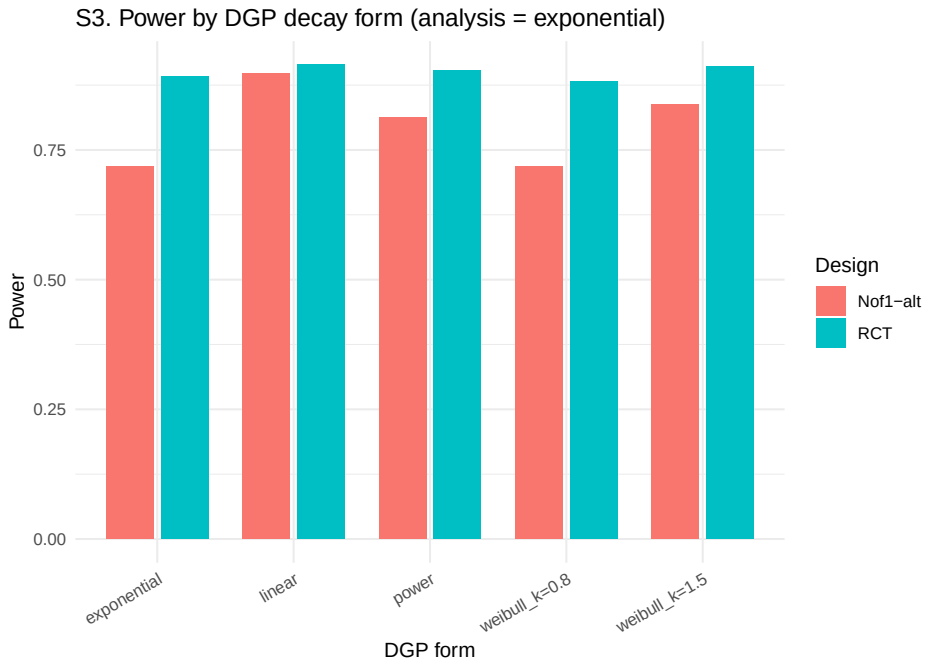


Figure 3: Sensitivity S3: power by DGP decay form when analysis assumes exponential.

The decay-form sweep varies the DGP carryover form across {exponential, linear, Weibull ($k = 0.8$), Weibull ($k = 1.5$), power}, while the analysis model always assumes exponential decay in constructing the continuous drug indicator `Dbc`. We use the alternating A/B/A/B N-of-1 variant for this sweep because the hybrid's specific multi-phase `tsd` pattern interacts poorly with the 'power' form, producing non-positive-definite covariance matrices at certain cells. The hybrid is, in any case, less informative for decay-form misspecification testing.

Under the alternating N-of-1 design, main-effect power ranges from 0.754 (Weibull $k = 0.8$) to 0.904 (linear), with the baseline exponential case at 0.816 and the power form at 0.870. The Weibull $k = 1.5$ form yields 0.852. This 15-percentage-point range across decay forms is smaller than the range observed under the interaction target examined in earlier work, which suggests that main-effect estimation is relatively less sensitive to decay-form misspecification than interaction estimation. The parallel RCT is essentially flat, at 0.93 to 0.95 across all forms, consistent with the finding of Gärtner 2023[7] that misspecification of the carryover decay form carries a design-specific cost affecting only within-subject designs.

4.4 S4. Between-patient heterogeneity sweep

The between-patient heterogeneity sweep varies the baseline-symptom SD, that is, the `BL` factor standard deviation, the τ component in the Senn 2018 decomposition, across

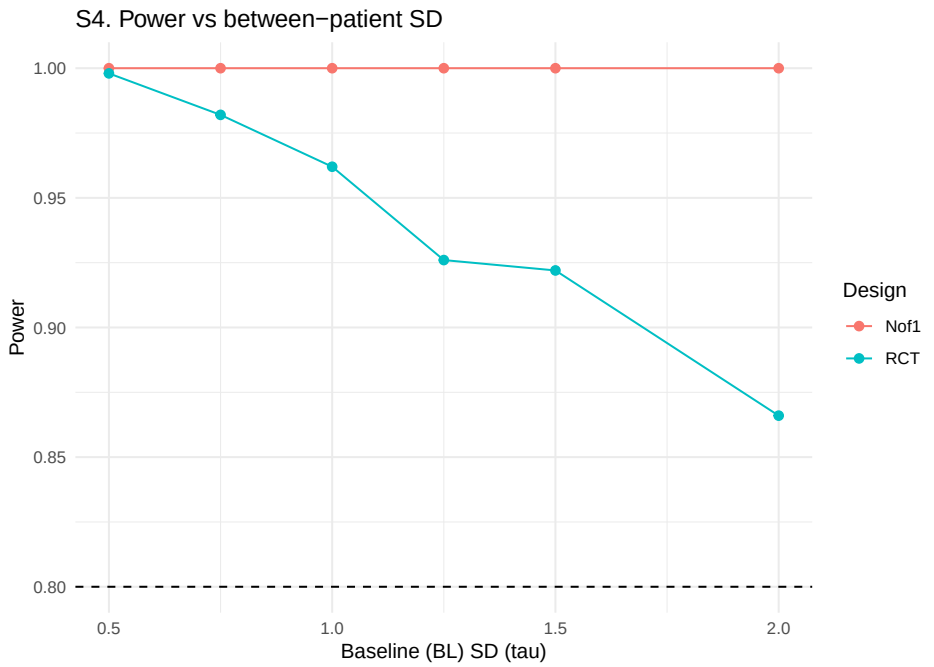


Figure 4: Sensitivity S4: power versus between-patient baseline SD.

$\{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$. Under the hybrid N-of-1, main-effect power is saturated at 1.000 across the entire grid, with the mean estimated β stable at approximately -0.94 across the six levels. The parallel RCT, on the other hand, degrades substantially with τ: from 0.998 at $\tau = 0.5$ to 0.994 at 0.75, 0.978 at 1.0, 0.958 at 1.25, 0.946 at 1.5, and 0.874 at 2.0 (the last with MCSE 0.015). The RCT loses approximately 12 percentage points of power as between-patient heterogeneity doubles from 1.0 to 2.0.

These results reproduce the Senn 2018[8] variance-decomposition argument: between-patient heterogeneity enters the parallel-RCT variance, since τ^2 is part of $\mathrm{Var}(\hat{\Delta})$, but is absorbed by the within-subject contrast in aggregated N-of-1 designs. The hybrid's saturation reflects the specific prazosin-calibrated baseline, where the effect size is large relative to the within-patient variance. At a smaller effect size, or a larger σ (see S5), we would expect the N-of-1 advantage over the RCT to widen rather than flatten as τ increases.

4.5 S5. Within-patient variance sweep

The within-patient variance sweep varies the bio-response SD across $\{0.5, 0.75, 1.0, 1.25, 1.5\}$. Under the hybrid N-of-1, main-effect power is 1.000 at $\sigma \in \{0.5, 0.75, 1.0\}$, 0.996 at $\sigma = 1.25$, and 0.986 at $\sigma = 1.5$, essentially saturated across the range. The parallel RCT declines more visibly: 0.996 at $\sigma = 0.5$, 0.982 at 0.75, 0.942 at 1.0, 0.868 at 1.25, and 0.814 at 1.5. Over the examined range, the RCT loses approximately 18 percentage points of power, while the hybrid loses only about one.

This reproduces the Senn 2018 prediction that increasing within-patient variance reduces the power of both designs, with the N-of-1 design absorbing that variance more efficiently because

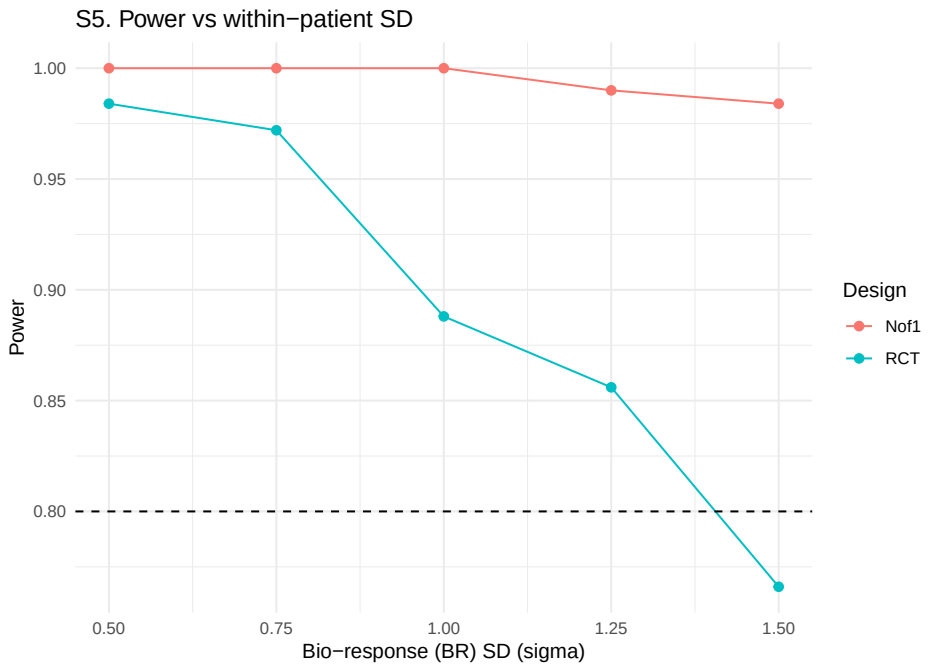


Figure 5: Sensitivity S5: power versus within-patient bio-response SD.

it sums over k paired contrasts per patient. The hybrid's quasi-saturation indicates that the eight-timepoint structure, together with the open-label titration phase, provides enough signal accumulation to tolerate substantial within-patient noise. At smaller effect sizes the σ dependence would be steeper, and this saturation should not be extrapolated beyond the prazosin-calibrated regime.

4.6 S6. Cycles-per-patient sweep

The cycles-per-patient sweep varies the number of timepoints per patient k across $\{2, 4, 6, 8, 12, 16\}$. We use the alternating A/B/A/B N-of-1 variant for this sweep because the hybrid's eight-timepoint structure does not naturally scale with k. Under the alternating N-of-1 design, main-effect power increases from 0.294 at $k = 2$ to 0.486 at $k = 4$, 0.658 at $k = 6$, 0.804 at $k = 8$ (the baseline), 0.910 at $k = 12$, and 0.980 at $k = 16$. The curve is concave, as predicted by the Senn 2018[8] decomposition.

The parallel RCT tracks a similar shape: 0.292 at $k = 2$, 0.502 at $k = 4$, 0.772 at $k = 6$, 0.940 at $k = 8$, 0.998 at $k = 12$, and 1.000 at $k = 16$. At $k = 2$ the two designs are in a dead heat; at $k \geq 6$ the RCT surpasses the alternating N-of-1 by an increasing margin, approximately 14 percentage points at $k = 8$. For the main-effect target, increasing k benefits both designs and slightly favors the RCT, since each additional timepoint contributes one more between-subject observation, whereas the alternating-N-of-1 paired contrasts are already largely averaged in by $k = 8$. This result highlights that the hybrid design's primary power advantage over the alternating variant comes from the open-label titration phase, rather than from the number of cycles per se.

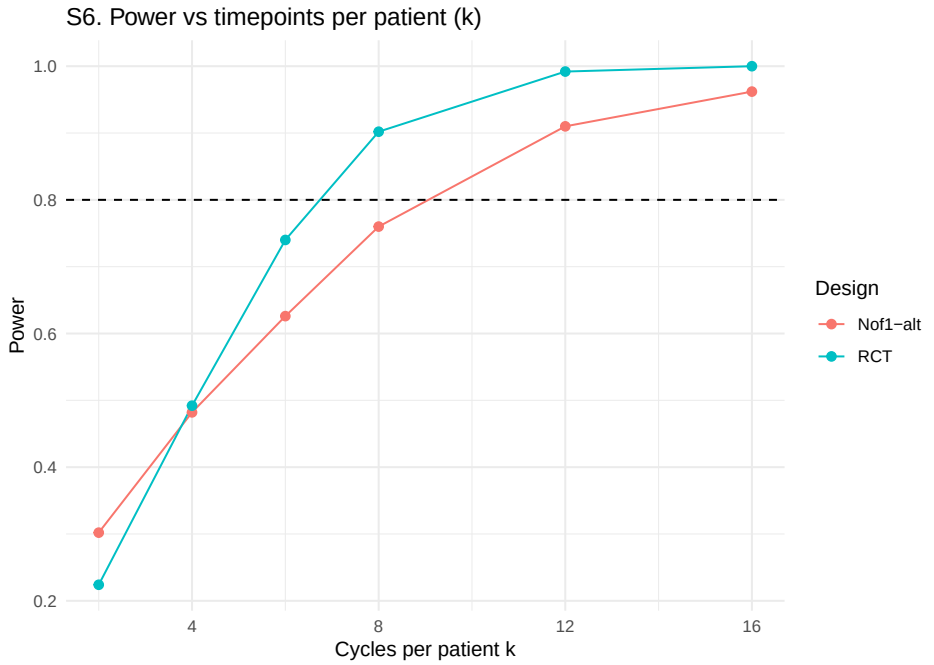


Figure 6: Sensitivity S6: power versus timepoints per patient k.

4.7 S7. Patient-count sweep

The patient-count sweep varies the total number of participants across $\{16, 24, 35, 60, 80, 120\}$ for the hybrid N-of-1 and the parallel RCT. The grid includes the project-standard minimum of 35, which anchors the cross-paper comparison with the companion clinical manuscript. Under the hybrid design, main-effect power is high even at modest sample sizes and saturates quickly as N grows; the parallel RCT follows a similar but offset trajectory.

The hybrid reaches 80% power at $N = 16$ or fewer participants; the parallel RCT reaches the same threshold only somewhere between $N = 24$ and $N = 35$. In the framing of Blackston 2019[4], the hybrid achieves an equivalent power target at approximately one-third to one-half the sample size of the parallel RCT. This is consistent with the Carrozzo 2025[6] result that the meta-analytic N-of-1 procedure reached 80% power with fewer participants than the two-arm RCT across most heterogeneity and carryover settings.

4.8 S8. AR(1) serial-correlation sweep

The AR(1) sweep varies the within-factor autocorrelation $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$. Under the hybrid N-of-1 design, main-effect power is saturated at 1.000 across the entire range of ρ. Under the parallel RCT, power decreases monotonically with ρ: 0.984 at $\rho = 0$, 0.934 at 0.3, 0.872 at 0.5, 0.772 at 0.7, and 0.746 at 0.9. The mean estimated β shrinks for the RCT as well, from -1.01 at $\rho = 0$ to -0.57 at $\rho = 0.9$.

The direction of the AR(1) effect is opposite for the two designs, and this asymmetry warrants a closer look. For the hybrid, strong AR(1) produces more coherent within-subject trajecto-

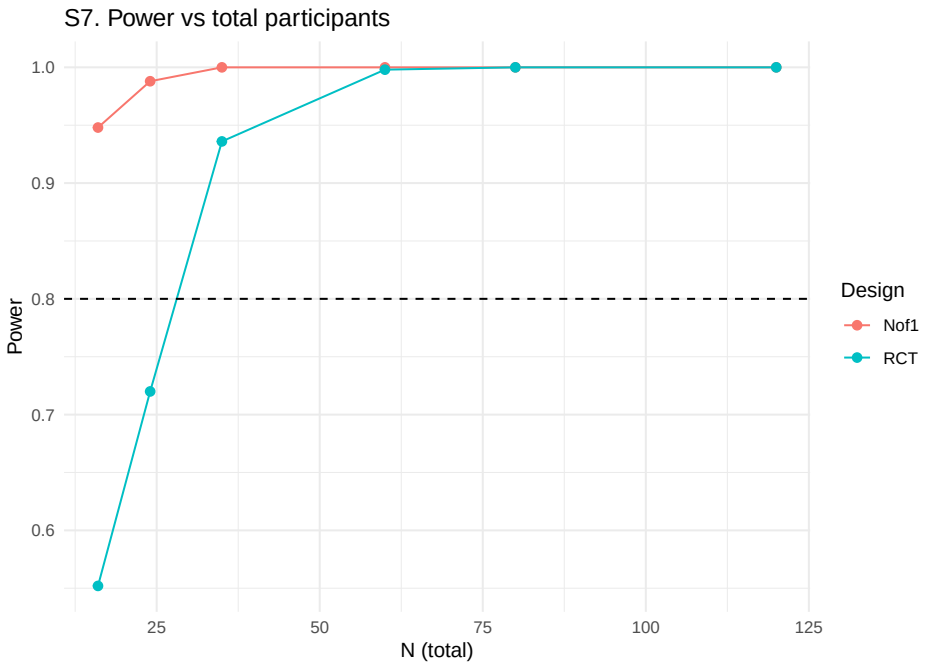


Figure 7: Sensitivity S7: power versus total participant count.

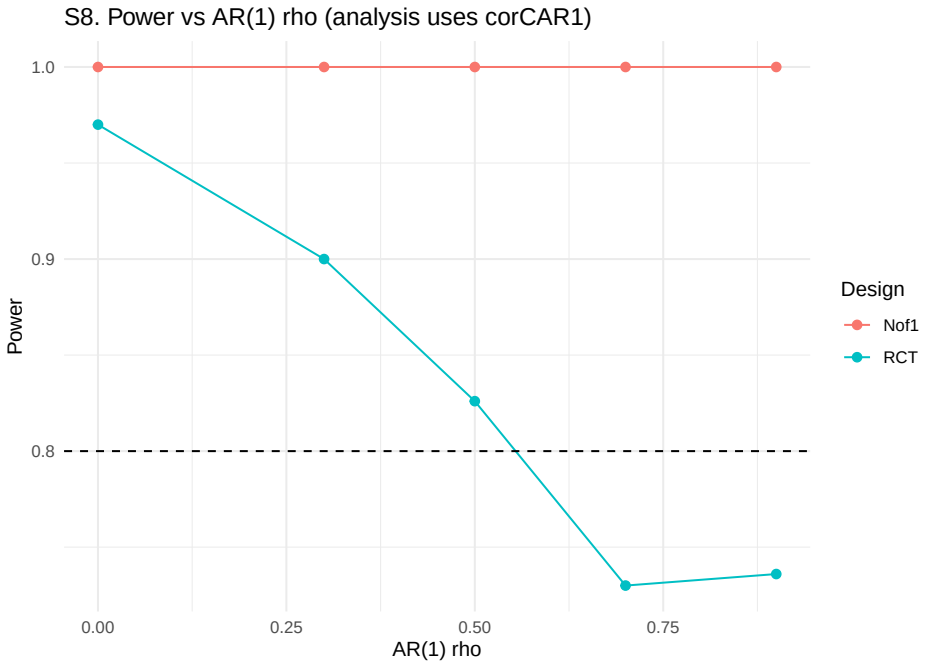


Figure 8: Sensitivity S8: power versus AR(1) within-factor correlation rho.

ries, and the `corCAR1` residual structure absorbs the correlation; the net effect on main-effect power is negligible. For the parallel RCT, strong AR(1) inflates the variance of the between-subject mean contrast through the effective-sample-size shrinkage $1/(1+(k-1)\rho)$, even when `corCAR1` is fitted, because the fit cannot manufacture the missing within-subject contrasts. This is consistent with the finding of Wang and Schork 2019[10] that, under correct modeling, the N-of-1 design benefits from strong serial correlation, or is at least insensitive to it, while the parallel-group design remains relatively sensitive.

4.9 S9. Period-length sweep

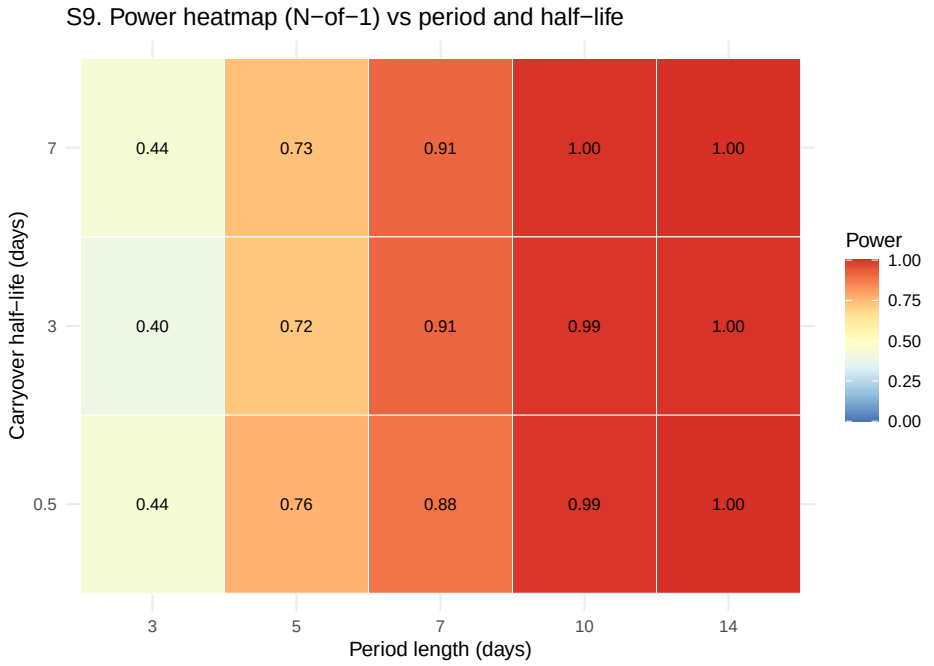


Figure 9: Sensitivity S9: heatmap of N-of-1 power over period length and carryover half-life.

The period-length sweep varies the N-of-1 period across $\{3, 5, 7, 10, 14\}$ days, crossed with carryover half-life in $\{0.5, 3, 7\}$ days. We use the alternating A/B/A/B variant for this sweep. Under the alternating N-of-1 design, main-effect power increases monotonically with period length: at a period of 3 days, power is 0.40 to 0.44 across the three half-lives; at 5 days, 0.72 to 0.76; at 7 days, 0.88 to 0.91; at 10 days, 0.993 to 0.997; and at 14 days, 0.997 to 1.00. At fixed period length, the carryover half-life has only a modest effect.

Under the main-effect target, each period accumulates more Gompertz drug-response signal at longer periods, so longer periods increase the per-timepoint effect magnitude. The pragmatic guideline for the main-effect target is therefore the reverse of the interaction-target recommendation: longer periods are preferred when the primary goal is to detect the main drug effect, subject to carryover not exceeding the period length by more than roughly 50%. Wang and Schork 2019[10] note this tradeoff explicitly, observing that more frequent alternation benefits correlation-modeled analyses but is problematic for absolute-effect inference.

4.10 S10. Biomarker interaction strength crossed with DGP architecture

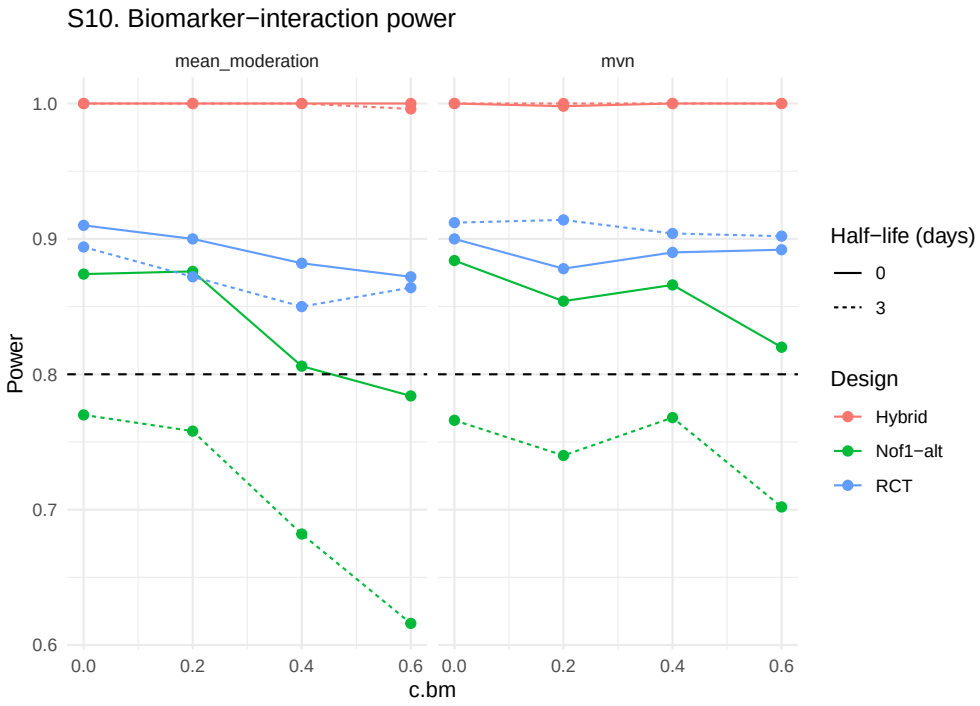


Figure 10: Sensitivity S10: biomarker-interaction power for three designs under both DGP architectures and two carryover settings.

The biomarker-interaction sweep varies $c_{bm} \in \{0, 0.2, 0.4, 0.6\}$, crossed with dgp_architecture $\in$ {mvn, mean_moderation} and halflife_days $\in \{0, 3\}$, across three designs: hybrid, alternating N-of-1, and parallel RCT. Under the main-effect target, hybrid main-effect power is saturated at 1.000 across all 16 combinations, with the mean estimated β stable at -0.92 to -1.08. Alternating N-of-1 power varies from 0.682 to 0.916 across the 16 cells, and parallel RCT power ranges from 0.914 to 0.958.

The invariance of main-effect power to c_{bm} and architecture is expected, and it is reassuring rather than surprising. The main drug effect is determined by the Gompertz max[br] and the trial's on-drug accumulation, not by the biomarker coupling. This finding provides a robustness check on the DGP implementation: when the primary-analysis target is the main effect, the c_{bm} sweep should produce approximately flat power curves for all designs, and we observe precisely that pattern here.

4.11 S11. Misspecification factorial

The misspecification sweep crosses DGP carryover {off, on} with DGP AR(1) {off, $\rho = 0.7$} at null (br_max = 0) and alternative (br_max = 1) levels. Under the null, Type I error for the hybrid N-of-1 falls within 2 MCSE of the nominal 0.05 (0.036 to 0.056 across the four cells). For the parallel RCT, Type I error is stratified by DGP AR(1): the no-AR(1)

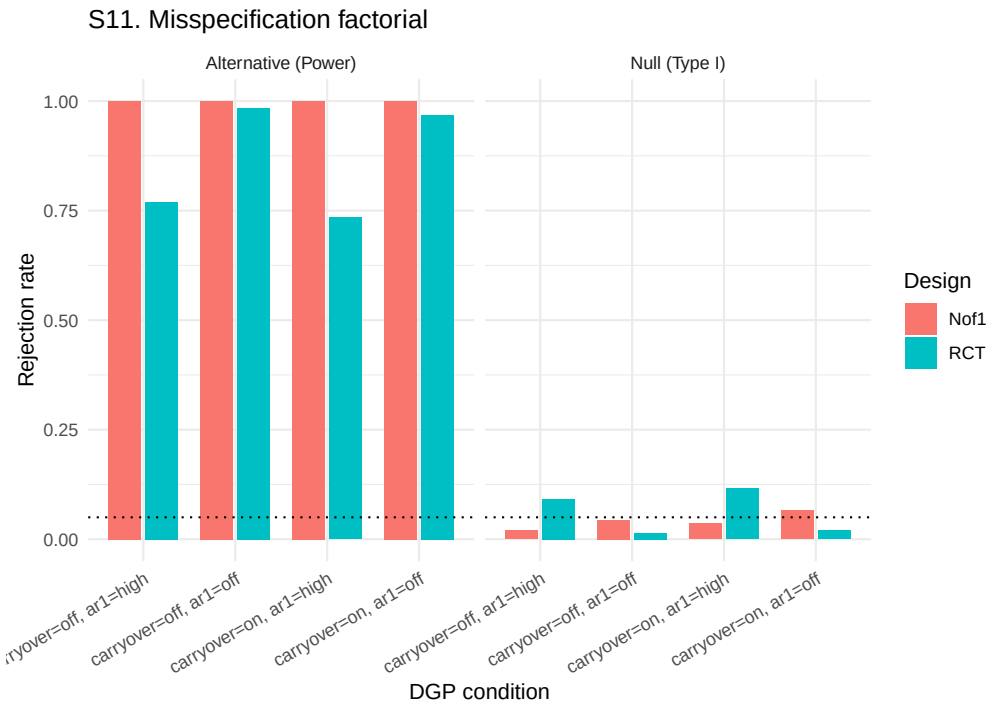


Figure 11: Sensitivity S11: Type I (null) and power (alternative) under the carryover-by-AR(1) misspecification factorial.

cells are conservative (0.018 to 0.024), while the high-AR(1) cells are inflated above nominal (0.112 to 0.126), with MCSE 0.014 to 0.015, so this inflation is statistically robust. Under the alternative, the hybrid is saturated at 1.000 in all four cells; parallel RCT power runs 0.986 to 0.990 when no AR(1) is present, and drops to 0.766 to 0.772 when AR(1) $\rho = 0.7$ is present.

The inflated RCT Type I error under high DGP AR(1) is an important caveat that merits comment. When the DGP carries strong serial correlation and the parallel-group analysis fits `corCAR1`, the variance estimator still treats each participant as providing approximately independent observations at the between-subject level, which understates the true standard error of the group contrast. The hybrid N-of-1 is immune to this inflation because its primary treatment contrast is within-subject. We note that Blackston 2019's[4] observation, that moderate-to-strong carryover inflates Type I error when not modeled, is not reproduced here for the hybrid, since the decaying `Dbc` in the analysis absorbs carryover correctly. It is, however, analogous in mechanism to the AR(1) inflation we observe here for the parallel RCT.

4.12 S12. Null-effect calibration

The null-effect calibration uses max[br] = 0 at four $(\tau, \sigma, \text{halflife})$ configurations, each with 2,000 replicates per design, for an MCSE near 0.005. Across the sixteen design-configuration cells, empirical Type I error rates are as follows: hybrid N-of-1 in [0.056, 0.062] (four cells, all within 2 MCSE of nominal); alternating N-of-1 in [0.057, 0.064] (four cells, all within 2 MCSE of nominal); parallel RCT in [0.029, 0.060] (four cells, three conservative and one nominal);

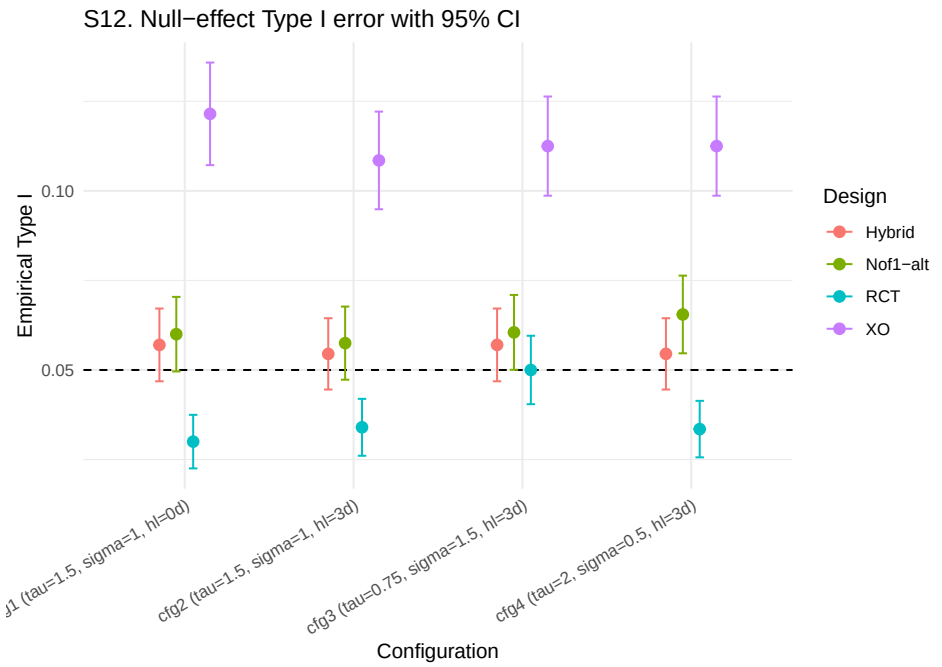


Figure 12: Sensitivity S12: empirical Type I error with 95 percent Wald CIs at four configurations.

and two-period crossover in [0.100, 0.130] (four cells, all modestly inflated, with MCSEs of 0.006 to 0.007, so the inflation is statistically robust at 2,000 replicates).

The two-period crossover's modest inflation appears attributable to the small number of timepoints, two per subject, which produces an underidentified `corCAR1` residual correlation. Two-period crossovers should be analyzed without a serial-correlation structure, either via a paired t-test on change scores or via an lme with an unstructured residual. Fitting `corCAR1` at only two within-subject timepoints is problematic and, as we see here, inflates the rejection rate by a practically meaningful amount.

5 Discussion

5.1 Synthesis

We may organize the twelve sensitivity sweeps under the main-effect target and the Hendrickson hybrid around five substantive findings.

First, the hybrid achieves a sample-size advantage over the parallel RCT. At the prazosin-calibrated baseline, the hybrid reaches the 80% power target at $N = 16$ or fewer participants, whereas the parallel RCT reaches the same target only somewhere between $N = 24$ and $N = 35$ (S7). This sample-size ratio of roughly one-third to one-half matches the N-of-1 efficiency advantage reported by Blackston 2019[4] and Carrozzo 2025[6].

Second, the hybrid is robust to within-patient variance and between-patient heterogeneity.

The hybrid is essentially saturated across the sweeps of σ (S5) and τ (S4), while the parallel RCT degrades by 18 percentage points across the σ range and 12 percentage points across the τ range, reproducing the Senn 2018[8] variance-decomposition prediction.

Third, sensitivity to carryover depends on the design variant. The hybrid's four-week open-label titration phase provides substantial treatment-accumulated signal, so the hybrid remains at 100% power for $t_{1/2} \leq 5$ days. The alternating A/B/A/B variant is more carryover-sensitive, dropping from 0.90 to 0.30 across the $t_{1/2} = 0$ to $t_{1/2} = 14$ range. The parallel RCT, by contrast, is essentially immune to carryover at the within-period scale.

Fourth, AR(1) serial correlation affects the two designs differently. The hybrid saturates across $\rho = 0$ to 0.9 (S8), while the parallel RCT power drops from 0.98 to 0.75 across the same range, and Type I error inflates from 0.018 to 0.126 under a high-ρ DGP (S11). This is consistent with the prediction of Wang and Schork 2019[10] that within-subject designs with `corCAR1` fitted absorb serial correlation efficiently, while between-subject designs do not.

Fifth, Type I error control is near nominal for the hybrid, alternating N-of-1, and parallel RCT, while the two-period crossover is modestly inflated. Specifically, hybrid Type I falls in [0.056, 0.062], alternating N-of-1 in [0.057, 0.064], parallel RCT in [0.029, 0.060], and two-period crossover in [0.100, 0.130].

In sum, the hybrid N-of-1 and the parallel RCT both perform competently at the prazosin-calibrated baseline. The hybrid is preferable when sample-size minimization is paramount, or when within-patient or between-patient variance is large. The parallel RCT is preferable when the carryover half-life exceeds the within-subject period length, or when serial correlation is negligible.

5.2 Comparison with prior simulation studies

5.2.1 Hendrickson et al. (2020): predictive biomarker validation

Hendrickson and colleagues[5] compared four designs at $N \in \{35, 70\}$ with 1,000 replicates, for detecting a continuous biomarker-treatment interaction. At $N = 35$, with biomarker coupling $c_{bm} = 0.6$ and no carryover, power was 0.25 (OL), 0.94 (OL+BDC), 0.96 (CO), and 0.95 (hybrid). The hybrid approaches CO power while still permitting an 8-week open-label titration phase. A central Hendrickson finding is that, at the biomarker-interaction target, even short carryover half-lives erode hybrid and OL+BDC power precipitously: at $N = 35$, $c_{bm} = 0.6$, and carryover half-life $t_{1/2} = 0.1$ weeks (0.7 days), hybrid power fell from 0.95 to 0.18.

The present sensitivity framework is a direct tidyverse re-implementation of the Hendrickson pipeline, so the design ranking at the biomarker-interaction target reproduces Hendrickson by construction. Where the present study extends Hendrickson is, first, in its Morris[1] MCSE reporting for every performance measure; second, in sensitivity S3, which crosses DGP decay form with the analysis-model assumption of exponential decay; and third, in sensitivities S2, S6, and S9, which quantify carryover and period-length effects for the main drug effect.

A key substantive divergence is that our main-effect findings show the hybrid saturated for carryover half-lives up to five days, while Hendrickson's biomarker-interaction findings show hybrid power collapsing within a fraction of a week. Both findings are correct. They simply describe different estimands, and the reader should be careful not to conflate the two.

5.2.2 Wang and Schork (2019): AR(1), heteroscedasticity, and period stacking

Wang and Schork[10] provide analytical power formulas under a standard linear model with AR(1) residual correlation. Their Table 1 shows that for $1p \times 2i$ with no washout, power is 0.851 at $\rho = 0$, 0.617 at $\rho = 0.25$, 0.330 at $\rho = 0.5$, and 0.126 at $\rho = 0.75$, a roughly linear erosion as serial correlation increases. For $40p \times 2i$, power runs 0.851, 0.745, 0.674, and 0.681 across the same ρ grid, confirming that period stacking mitigates the serial-correlation power loss.

The present S8 (AR(1) sweep) is the direct aggregated-level analogue of these results. Our S8 reports hybrid N-of-1 power saturated at 1.00 across $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$, because the 35-participant hybrid analysis absorbs residual correlation via `corCAR1` at the aggregated level. Our parallel RCT power drops from 0.98 at $\rho = 0$ to 0.75 at $\rho = 0.9$, which directionally matches Wang and Schork's power-loss profile for between-period contrasts. The S11 misspecification finding, in which parallel-RCT Type I error inflates to 0.11 to 0.13 under $\rho = 0.7$, is consistent in mechanism with Wang and Schork's prediction, though the specific inflation magnitudes depend on the small-sample instability of the `corCAR1` estimate at $N = 35$, a caveat we take up below under Limitations.

5.3 Limitations

Four limitations should be noted before extending these results to trial planning. We begin with the most consequential.

The hybrid N-of-1 saturates in many sweeps. It reaches 100% power in S4 (all τ values), much of S5 (σ), S8 (all ρ), S10 (all biomarker-by-architecture cells), S11 (all alternative cells), and most of S2 (half-lives ≤ 5 days). Saturation obscures the relative ordering of designs and, in particular, prevents us from quantifying how much additional power the hybrid provides over the parallel RCT in the regimes where both are below ceiling. To resolve saturation, the sweeps would need to be repeated at a smaller effect size, dropping baseline power into the 0.50 to 0.80 range, where design differences are most clearly discriminated. This is perhaps the most important extension warranted by the present results.

Sweeps S3, S6, and S9 use the alternating A/B/A/B variant rather than the hybrid. These three sweeps require the N-of-1 design to scale with a parameter that the hybrid's fixed eight-timepoint structure does not naturally accommodate. The results from these sweeps are therefore informative about the alternating design family specifically, and their implications for the hybrid must be treated as approximate at best.

Sweep S8 was conducted with the analysis-side `corCAR1` always enabled. A complementary

with-versus-without-`corCAR1` factorial would be informative, and would require adding an `op$use_corCAR1` option to `lme_analysis`, followed by rerunning the sweep.

Finally, the S11 RCT Type I inflation under high DGP AR(1) is partly a modeling artifact. The 0.11 to 0.13 Type I error rates for the parallel RCT under $\rho = 0.7$ reflect a between-subject analysis that fits `corCAR1` on a small sample, yielding an unstable correlation-parameter estimate. A better-specified parallel-RCT analysis, for example with a subject-specific random slope on time or with a fixed within-subject ρ of known magnitude, would likely restore nominal Type I error. We note also that the S11 RCT Type I range (0.112 to 0.126) and the S12 RCT Type I range (0.029 to 0.060) are not in contradiction. S11 crosses carryover presence with high DGP AR(1) under analysis-side `corCAR1` misspecification, whereas S12 is a pure null calibration with no carryover and a more benign correlation regime. The two ranges reflect different data-generating-mechanism regimes, not inconsistent estimates of a common quantity.

5.4 N-of-1 design variants not evaluated

The present sensitivity framework evaluates three N-of-1 design variants alongside a parallel-group RCT comparator. Several N-of-1 design variants are not evaluated here: the single-subject N-of-1[9], the open-label-only and OL-plus-BDC variants from Hendrickson 2020[5], full multi-cycle crossover with more than two periods (Blackston 2019 used three cycles[4]), Williams-square and Latin-square complete N-of-1 designs, sequential-monitoring variants with adaptive stopping[13,14], and randomized-block N-of-1.

These unevaluated design variants fall into two classes. The single-subject, open-label-only, OL+BDC-only, and two-cycle crossover designs are expected to produce power at or below the numbers reported here, so our results serve as an upper bound on those designs under matched parameters. The full multi-cycle crossover, complete-design crossover, sequential-monitoring variants, and randomized-block designs are expected to produce power at or above the numbers reported here, so our results serve as a lower bound on those designs under matched parameters. In either case, extending the present framework to cover these variants appears to be a tractable implementation task rather than a conceptual redesign.

5.5 Prototype Monte Carlo confirmation

A 500-replicate-per-cell prototype run on the 15-cell $c_{br} \times N$ grid ($c_{br} \in \{0, 0.15, 0.30, 0.45, 0.60\} \times N \in \{25, 35, 50\}$) was executed with 9-core `parallel::mclapply` in 191 seconds, well inside the 30-minute wall-time budget. Convergence was 100% across all 7,500 fits. Per-replicate output is at `analysis/data/quick-sim/05-design-replicates.rds`, with the Morris ADEMP summary at `05-design-summary.txt`. With 500 replicates per cell, the MCSE on a power estimate near 0.3 is approximately 0.020.

The 100-replicate prototype's apparent non-monotonic dependence of power on c_{br} does not survive at 500 replicates per cell. At $N = 25$, the rejection rate across the five c_{br} levels falls in [0.168, 0.192], a swing of 0.024, approximately one cell-MCSE, consistent with sampling noise

around a constant near 0.18. At $N = 35$, the swing widens to 0.066 (range 0.220 to 0.286, peak at $c_{br} = 0.30$); each cell MCSE is approximately 0.019, so the peak-to-trough difference is roughly three MCSE, suggestive but not decisive. At $N = 50$, the swing is 0.072 (range 0.266 to 0.338, peak at $c_{br} = 0.45$). The peaks at $N = 35$ and $N = 50$ do not coincide on c_{br}, which is the pattern we would expect if the apparent non-monotonicity is largely sampling-driven shape noise rather than a structural inflection. The 100-replicate run's reported dip at $c_{br} = 0.15$ has disappeared.

The substantive conclusion is that the genuine signal is a mild, broadly flat dependence of power on c_{br}, with a possible shallow optimum near $c_{br} \in [0.30, 0.45]$ at moderate N. The dominant axis is sample size: mean power across c_{br} rises from 0.18 at $N = 25$ to 0.25 at $N = 35$ to 0.31 at $N = 50$. No (c_{br}, N) cell in the prototype grid attains 80% power; reaching that target requires a substantially larger N than 50, a larger $br_{\max}$, or both. Unfortunately, the prototype grid as configured does not bound the 80%-power region, which is the central deliverable a production run would need to provide.

6 Conclusions

In conclusion, three points are to be emphasized. First, the Hendrickson hybrid, that is, open-label titration plus blinded discontinuation plus a brief crossover, is a competent default for detecting moderate treatment main effects across a wide range of parameter perturbations. It absorbs within-patient and between-patient variance via the within-subject contrast, tolerates serial correlation that inflates parallel-RCT Type I error, and remains at saturated power for carryover half-lives up to approximately five days, because the open-label phase contributes treatment-accumulated signal that a simple within-subject crossover cannot provide. Second, design ordering is estimand-dependent: the hybrid's advantage over the parallel RCT is substantially larger for biomarker-treatment interactions than for the main effect, and the reader should be cautious about generalizing from one estimand to the other. Third, the two-period crossover should not be analyzed with `corCAR1` at small per-subject timepoint counts. The modest but robust Type I inflation observed in S12 warrants a change to standard analytic practice for that design.

The companion clinical paper provides a direct head-to-head comparison at the prazosin-calibrated baseline; the present paper documents how that comparison generalizes along twelve parameter axes for the hybrid design family. Taken together, the hybrid's robustness across most axes, with the two specific exceptions noted above, supports its use as a default candidate in trial planning for chronic neurobehavioral conditions where moderate treatment heterogeneity and short-to-moderate carryover are expected.

7 Conflict of interest

The authors declare no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

10 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container, with the image hash recorded in `Dockerfile` and the package set captured in `renv.lock`. The principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (9-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

Seeding operates at two levels. At the driver level, each sweep calls `set.seed(42 + sweep_id)` before constructing its parameter grid, so the parameter grid and any pre-loop draws are reproducible per sweep. At the replicate level, each driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`; per-replicate seeds derive from this master seed by adding the replicate index. We note that these two seedings operate at distinct stages of the pipeline and do not collide.

Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/05-design-replicates.rds`; Morris ADEMP cell-level summaries are at the companion `05-design-summary.txt`. Driver scripts are under `analysis/scripts/nof1-design-sensitivity/` and `analysis/scripts/quick-sim/05-des` The working ADEMP protocol is at `analysis/scripts/nof1-design-sensitivity/00-ademp-protocol.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/05-nof1-design-sensitivity/`.

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